

Set 32: Solution concentrations

Notes

Many reactions occur in aqueous solutions. Stoichiometry in solutions uses the concept of concentration.

Concentration is defined as the amount of solute per unit volume of solution. The amount of solute is usually expressed in moles. The relationship can be expressed as:

$$n = cV$$

where **n** is the number of moles of solute (mol),

c is the concentration of solute in moles per litre (mol L⁻¹), and

V is the volume of the solution in litres (L).

Note: Solution concentrations can be expressed using 'c' or '[]' that is, c(NaCl) or [NaCl] both refer to the concentration of NaCl in solution, expressed in mol L⁻¹.

Examples

1. Find the number of moles of sodium chloride in 15.0 mL of a 2.50 mol L⁻¹ NaCl solution.

$$\begin{aligned} n &= cV \\ &= (2.50)(15.0 \times 10^{-3}) \\ n &= 3.75 \times 10^{-2} \text{ mol of NaCl} \end{aligned}$$

2. Calculate the concentration in mol L⁻¹ of an iron(II) sulfate-7-water solution which contains 5.56 g of the solute dissolved in 750 mL of solution.

(a) Calculate the number of moles of solute:

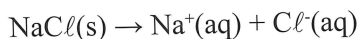
$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{5.56}{278.0} \quad M(\text{FeSO}_4 \cdot 7\text{H}_2\text{O}) = 278.0 \text{ g mol}^{-1} \\ &= 2.00 \times 10^{-2} \text{ mol} \end{aligned}$$

(b) Calculate the concentration of FeSO₄·7H₂O solution.

$$\begin{aligned} c &= \frac{n}{V} \\ &= \frac{2.00 \times 10^{-2}}{750 \times 10^{-3}} \\ c &= 2.67 \times 10^{-2} \text{ mol L}^{-1} \end{aligned}$$

Concentration of ions

Ionic substances and some covalent substances, such as strong acids, produce almost 100% ions when dissolved in H₂O. For example



Knowing the concentration of the solute and its formula, the concentration of ions formed in solution can be calculated.

3. In 25.0 mL of a 0.200 mol L⁻¹ solution of aluminium sulfate, calculate:
- (a) the [Al³⁺] (c) n(Al³⁺)
 (b) the [SO₄²⁻] (d) n(SO₄²⁻)
- (i) Write a balanced ionic equation:
 $\text{Al}_2(\text{SO}_4)_3(\text{s}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{SO}_4^{2-}(\text{aq})$
- (ii) From the balanced ionic equation relate the unknowns to the known:
- (a) $\text{c}(\text{Al}^{3+}) = 2 \times \text{c}(\text{Al}_2(\text{SO}_4)_3)$
 $= (2) (0.200)$
 $\text{c}(\text{Al}^{3+}) = 0.400 \text{ mol L}^{-1}$
- (b) $\text{c}(\text{SO}_4^{2-}) = 3 \times \text{c}(\text{Al}_2(\text{SO}_4)_3)$
 $= (3) (0.200)$
 $\text{c}(\text{SO}_4^{2-}) = 0.600 \text{ mol L}^{-1}$
- (c)(i) Calculate the number of moles of Al₂(SO₄)₃
 $\text{n}(\text{Al}_2(\text{SO}_4)_3) = \text{c}(\text{Al}_2(\text{SO}_4)_3) V$
 $= (0.200) (25.0 \times 10^{-3})$
 $= 5.00 \times 10^{-3} \text{ mol}$
- (ii) Relate the unknown numbers of moles to the known number of moles of Al₂(SO₄)₃:
- $\text{n}(\text{Al}^{3+}) = 2 \times \text{n}(\text{Al}_2(\text{SO}_4)_3)$
 $= (2) (5.00 \times 10^{-3})$
 $\text{n}(\text{Al}^{3+}) = 1.00 \times 10^{-2} \text{ mol}$
- (d) $\text{n}(\text{SO}_4^{2-}) = 3 \times \text{n}(\text{Al}_2(\text{SO}_4)_3)$
 $= (3) (5.00 \times 10^{-3})$
 $\text{n}(\text{SO}_4^{2-}) = 1.50 \times 10^{-2} \text{ mol}$

Set 32: Exercises

- Calculate the concentrations of the following solutions:
 - 0.223 moles of copper(II) sulfate in 125.0 mL of solution
 - 1.17 moles of sodium chloride in 2.05 L of solution
 - 0.0335 moles of silver nitrate in 250.0 mL of solution
- Calculate the number of moles of:
 - potassium nitrate in 105 mL of 2.55 mol L⁻¹ potassium nitrate solution
 - sodium carbonate in 2.50 L of 0.112 mol L⁻¹ of sodium carbonate solution
 - potassium permanganate in 660 mL of 0.230 mol L⁻¹ of potassium permanganate solution

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3. Calculate the mass of solute that must be used in order to prepare each of the solutions listed below:
 - (a) 630 mL of 1.26 mol L⁻¹ KCl solution from KCl
 - (b) 250 mL of 0.265 mol L⁻¹ Na₂CO₃ solution from Na₂CO₃·10H₂O
 - (c) 1.05 L of 0.420 mol L⁻¹ H₂C₂O₄ solution from H₂C₂O₄·2H₂O
4. Calculate the number of moles of:
 - (a) chloride ions in 25.0 mL of 0.200 mol L⁻¹ barium chloride solution
 - (b) sulfate ions in 550 mL of 2.56 mol L⁻¹ sodium sulfate solution
 - (c) nitrate ions in 2.20 L of 2.02 × 10⁻³ mol L⁻¹ lead(II) nitrate solution
5. In a solution prepared by dissolving 10.0 g of potassium carbonate in 220 mL of water, determine:
 - (a) the concentration, in mol L⁻¹, of K⁺ (aq)
 - (b) the concentration, in mol L⁻¹, of CO₃²⁻ (aq)
6. Calculate the volume of 4.0 mol L⁻¹ nitric acid solution required to prepare 250 mL of 0.250 mol L⁻¹ solution.
7. Calculate the concentration in mol L⁻¹ of ammonium ions in a solution prepared by mixing 360 mL of 0.250 mol L⁻¹ ammonium sulfate solution with 675.0 mL of 1.20 mol L⁻¹ ammonium nitrate solution. Assume solution volumes are additive.
8. A laboratory assistant has a solution of 0.120 mol L⁻¹ KMnO₄. What volume of this solution must be diluted to produce 500 mL of a 0.025 mol L⁻¹ solution?
9. What volume of water must be added to 150 mL of 1.10 mol L⁻¹ sulfuric acid solution to prepare a 0.210 mol L⁻¹ solution?
10. 25.6 g of anhydrous sodium carbonate is dissolved in 200 mL of water.
 - (a) Calculate the concentration in mol L⁻¹ of this solution. Assume no volume change.
 - (b) A 20.0 mL sample of this solution was placed in a 150 mL flask and 80.0 mL of water added. Determine the concentration of sodium ions in this diluted solution.